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PLASMO: a simulation model for control of *Plasmopara viticola* on grapevine[1]

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The problem of protecting grapevine against diseases is an old one, but in the last few years new techniques have been developed to reduce cost to the farmer and damage to the ecosystem. These are based on mathematical models describing the state of the plant-parasite environment system. A model for forecasting development of grapevine downy mildew (*Plasmopara viticola*) is presented. The input variables are temperature, rainfall and leaf wetness (determining infection by sporangia), and RH and temperature (for incubation period). The model also takes into account the limited survival of spores. The output is expressed as % disease progress. Field validation tests, performed in 1990, 1991 and 1992 in several vineyards in Toscana (central Italy) showed a good correlation between observed and simulated infections. The model allowed the number of treatments to be reduced without any increase in downy mildew damage. It could in future be integrated with grapevine growth and development simulation models in an expert system to determine infected tissue area and thus the economic damage threshold.

Introduction

In Toscana (central Italy), downy mildew (*Plasmopara viticola*) is still the most destructive foliage disease of grapevine and protection measures represent 15–20% of the total cost of grapevine management. The traditional method of protection is to apply chemicals at fixed intervals (7–14 days) in order to maintain the tissue constantly covered by fungicide. Generally 8–12 treatments are required during one growing season. These applications are often unnecessary or performed in the total absence of disease. By forecasting the main phases of the life cycle of *P. viticola*, it is possible to determine when grapevine is susceptible to the fungus and when the fungus is exposed to chemicals. Treatments have to be applied at such times to obtain optimal control. Generally these times coincide with the end of the incubation period, when the sporangia are dispersed within the canopy onto healthy tissues. Each sporulation should be neutralized, because it can damage a large part of the vineyard, producing the inoculum for subsequent infections. Good timing of fungicide applications allows a considerable reduction in cost for vineyard managers and in chemical hazard to the environment. This method can be integrated with agronomical, mechanical and physical techniques, further reducing the number of chemical applications required.

The possibility of predicting downy mildew development is thus an important goal for epidemiological research, aiming to correlate pathogen development with environmental parameters and host physiology. These factors represent the whole of the plant-parasite environment system (Vanderplank, 1963), whose analysis allows mathematical relationships to

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be defined between its different components (Campbell & Madden, 1989). These relations can be used in models for simulating and forecasting the development of grapevine downy mildew, on the basis of biological and environmental inputs (Orlandini *et al.*, 1991).

In this paper, the life cycle of *P. viticola* is discussed and a simulation model is presented. Preliminary reports have been published and the model has been called PLASMO (Plasmopara Simulation Model) (Rosa, 1988; Rosa *et al.*, 1993). Starting from climatic variables, this model estimates development of the incubation period and other main life phases in % terms. Using this model, it is possible to time treatments better, to reduce the risk of fungicide tolerance in the pathogen and to achieve a better comprehension of downy mildew epidemiology (Magarey *et al.*, 1991). The input data needed can be manually inserted or recorded by an electronic data-acquisition system and thus automatically transferred and processed by personal computer. The proposed model has been validated in field experiments conducted for three seasons in the Chianti area.

Development of the model

Studies on the epidemiology of grapevine downy mildew (Müller & Sleumer, 1934; Goidanich *et al.*, 1958; Blaeser & Weltzien, 1979; Govi, 1983; Agrios, 1988; Lalancette *et al.*, 1988; Magarey & Weir, 1988; Hill, 1991; Magarey *et al.*, 1991) have shown that the different life phases of *P. viticola* develop according to climatic factors. In spring, the oospores germinate on the ground when soil wetness and temperature conditions are suitable. Primary infections (initiated from oospores) can occur at any time during the growing season, initiated by rainfall $\geqslant 8$ mm within any 24-h period after the minimum temperature is consistently higher than 10°C (Goidanich, 1959). The sporangia are dispersed by wind or rain splash. Infection by zoospores released from sporangia on the leaf surface only occurs at suitable leaf wetness and temperature (Wallin, 1967; Blaeser & Weltzien, 1979; Weltzien, 1981). Since zoospores penetrate through stomata, infection will only occur on grapevine shoots which have reached 100 mm in length. A growth simulation model can usefully replace the empirical rule (Miglietta *et al.*, 1992). The model simulates the duration of leaf wetness and the temperature necessary for infection using the hyperbolic function $f_1(T_t)$:

$$f_1(T_t) = \begin{cases} n/T_t & 6°C \leqslant T_t \leqslant 26°C \\ 0 & \text{otherwise} \end{cases}$$

where T_t = temperature at instant t (°C) and n = rate of increase of inoculum development. This function has been derived on the basis of data from Blaeser & Weltzien (1979) who consider the temperature sum required during the leaf wetness period. The hyperbolic function most representative of Blaeser & Weltzien's data was found by the least squares method. For simplicity, the function has been re-expressed as % infection progress $f_2(T_t)$:

$$f_2(T_t) = 100/f_1(T_t).$$

A result of 100% then means that leaves have been continuously wet for long enough to permit zoospore germination and growth of germ tubes through stomata.

The time from infection to formation of new sporangia (incubation period) varies from 4 to 25 days, strongly depending on temperature and RH. If the leaves can be covered with fungicide before the end of this period, later infection can be prevented. The duration of the incubation period (x_t) is computed by adding hourly increments, deduced from the temperature and RH for each hour. Each increment is expressed in terms of % progress at instant t and is indicated as d_t. A value of 100% thus indicates the end of the incubation period. If the temperature lies outside the range 'minimum temperature'—'maximum temperature', or if RH $< 30\%$, null increments occur. Inside the range of temperature, the increment is higher at 'optimal temperature', and

lower otherwise (function $f_3(T_t)$). Above RH 30%, the increment constantly increases (function $f_4(RH_t)$). The optimal temperature was observed to be about 22°C and the maximum temperature about 34°C (Goidanich, 1959). In the same study, as well as in another study (Magarey *et al.*, 1991), the minimum temperature was found to be about 10°C. The final result is a combination of these two effects, that is a function $f_5(T_t, RH_t)$ obtained as a multiplication of the two proposed functions $f_3(T_t)$ and $f_4(RH_t)$. A simple model could be proposed to determine the development of incubation period in % terms:

$$x_o = 0 \qquad \text{and} \qquad x_{t+1} = x_t + d_t$$

where x_t = % progress of incubation at instant t, d_t = hourly increment of % incubation at instant t,

$$d_t = f_5(T_t, RH_t)$$

$$f_5(T_t, RH_t) = f_3(T_t) * f_4(RH_t)$$

$$f_3(T_t) = 4*((T_t - Tmin)*(Tmax - T_t))/((Tmax - Tmin)^2)$$

$$f_4(RH_t) = m(RH_t - RHmin)$$

where RH_t = RH at instant t (%), RHmin = minimum RH (%), Tmin = minimum temperature (°C), Tmax = maximum temperature (°C), m = rate of increase of incubation development.

The produced sporangia can give rise to secondary infections, under the same condition of leaf wetness and temperature $f_2(T_t)$ as the primary infections, or can be neutralized by plant protection products. Adverse climatic conditions can also kill the sporangia, so preventing new infections. Thus a simple model simulates the limited survival of sporangia on the basis of temperature and humidity (Blaeser & Weltzien, 1979). The equation $f_6(T_t, RH_t)$ can be expressed in terms of % progress in dying off of the sporangia:

$$f_6(T_t, RH_t) = 100/f_7(T_t, RH_t)$$

$$f_7(T_t, RH_t) = ((f_8(T_t) - 12)*RH_t/100) + 12$$

$$f_8(T_t) = \begin{cases} 235.8 & T_t < 10°C \\ 1.8*T_t + 217.8 & 10°C \leqslant T_t \leqslant 15°C \\ 15.52*T_t + 12 & 15°C < T_t \leqslant 30°C \\ 12 & T_t > 30°C \end{cases}$$

Any infection, either primary or secondary, which is not neutralized, can give rise to new infections throughout the growing season. These start and follow the same dynamics as the previous ones. At the end of the season, the mycelium in the old leaves produces the wintering oospores.

On the basis of these considerations a simulation model (Fig. 1) has been devised. Optimal model parameters (m = 0.097 and n = 75.69) were chosen when the difference in time between observed and calculated sporangia emergences reached a minimum value. These parameters can be modified to adapt this model to different cultivars, environments, pathogen species, etc. The input variables of the model, with time step of 1 h, are air temperature and RH, rainfall and leaf wetness. Output variables are the number of *P. viticola* infections and, for each of them, % progress through the successive development phases (infection, incubation, spore survival, etc.). Outputs and inputs can be shown in numeric or graphic displays, to facilitate the use of the model by farm technicians. Moreover, the model considers the characteristics (persistence, efficiency, etc.) of treatments applied, so as to avoid new applications if the grapevines are still covered by an earlier one.

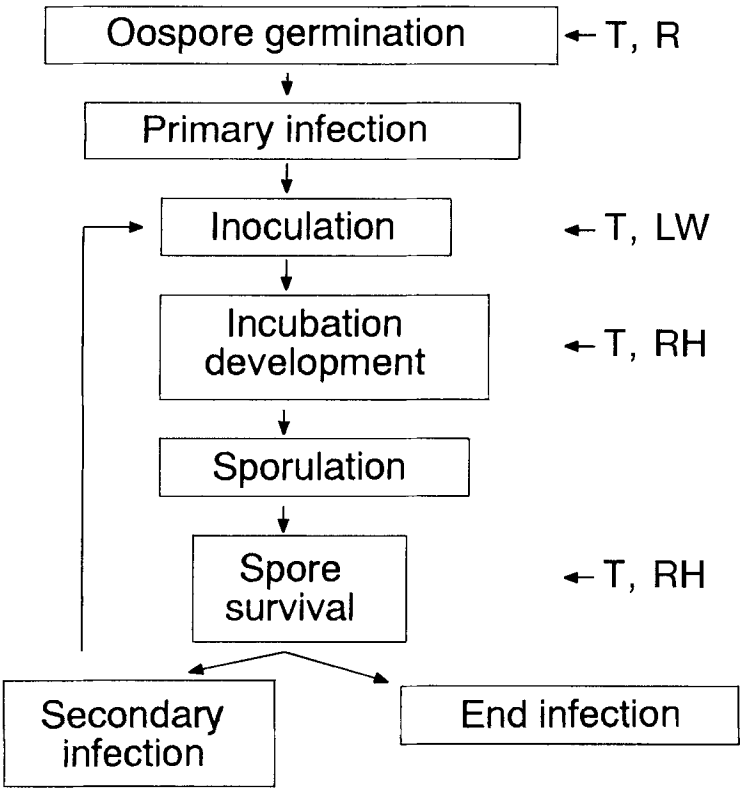


Fig. 1. Flow chart of PLASMO model. T = temperature; R = rainfall; LW = leaf wetness; RH = relative humidity.
Organigramme du modèle PLASMO.

Experimental

A 3-year (1990/1992) validation experiment was conducted at the Mondeggi-Lappeggi experimental farm (Firenze, central Italy), with the kind agreement of the Provincial Government of Firenze. The farm shows nearly the same geomorphological characteristics as those of the Chianti Classico region, even though it does not belong exactly to this particular area. Two vineyards were used, Pulizzano every year and Paretaio in 1990. They have an average altitude above sea level of 140–150 m, a slope of 14–18% and a prevailing northern (Pulizzano) or southern (Paretaio) aspect. The typical Chianti cultivars (Sangiovese, Canaiolo nero, Malvasia lunga del Chianti, Trebbiano toscano) are grown in the vineyards. Meteorological variables (temperature, RH, rainfall and leaf wetness) were recorded over three seasons in the centre of each vineyard.

The model was validated in two different phases: (1) verification of the simulation of *P. viticola* infections, (2) application in a disease management programme in the vineyard. The first step was to compare sporulation as observed in vineyards and as simulated by the model. Vineyard plots (1500 m^2) were left untreated with fungicide throughout the growing season to observe actual downy-mildew development. Epidemiological observations were made once a week on cv. Sangiovese. PLASMO was then used to determine fungicide application times in three vineyard plots (1500 m^2). The number of fungicide treatments and the disease intensity (DI) obtained in PLASMO plots were compared with those in three vineyard plots treated at fixed intervals (as usual for the Chianti region) and in three untreated plots. DI was determined throughout the

Table 1. Field key for determination of downy-mildew intensity on grapevine leaves
Echelle pour estimation au champ de l'intensité de mildiou sur feuilles de vigne

Class	Percentage	Description
1	0	No disease observed
2	1–25	A few oilspots only
3	26–50	Plants retain normal form, but leaves show many oilspots
4	51–75	Leaf area is destroyed and looks brown
5	76–100	Few leaves are green

growing season on leaves, the main organ damaged by downy mildew. In each vineyard plot, 200 leaves were observed to determine the surface damaged by downy mildew. The estimators' ability was evaluated and standardized and a field key (Kranz, 1987) was prepared to facilitate identification of downy-mildew symptoms in the vineyard (Table 1). The number and range of the classes were chosen to allow sufficient accuracy. Disease intensity was obtained as the average of values of each class weighted by the sample frequencies. The data was analysed by analysis of variance.

Results and discussion

The first phase of validation showed that the model fits correctly (Table 2). The number of simulated infections was higher in every year than the observed infections. However, all the observed infections were simulated by the model. Differences up to 5 days (Table 2) were observed in the date of sporangial emergence. These could be due to various factors, such as the difficulties in leaf-wetness measurement (Barr & Gillespie, 1987), in forecasting the date of oospore maturity (Tran Manh Sung *et al.*, 1990), in simulation of the duration of the incubation and latent period (Di Graziano *et al.*, 1988).

The second validation tests confirmed that the model allows a consistent reduction in the number of fungicide applications without increasing the virulence of downy-mildew infections (Table 3). Consistent differences in topoclimatic conditions among the years and vineyards considered did not affect performance of the model. Systemic or surface fungicides and application times were chosen in PLASMO plots on the basis of all developing infections. For

Table 2. Accuracy in timing of simulated downy-mildew infections by comparison with observed infections in Paretaio (PARE) and Pulizzano (PULI) vineyards in 1990, 1991 and 1992
Décalage entre les périodes simulées d'infection par le mildiou et les périodes observées dans les vignobles de Paretaio (PARE) et de Pulizzano (PULI) en 1990, 1991 et 1992

	PARE 90	PULI 90	PULI 91	PULI 92
Infection periods coincide	4	3	2	4
Simulation < 2 days late		1	1	2
Simulation < 5 days late				1
Simulation < 2 days early		1		
Simulation < 5 days early	1	1		1
No infection period observed	2	1	1	2
Total	7	7	4	10

Table 3. Comparison between number of treatments and downy-mildew intensity in PLASMO, fixed-interval and untreated plots during the validation tests
Comparaison du nombre de traitements et de l'intensité de mildiou dans les parcelles traitées selon le modèle PLASMO, selon un programme fixe, ou non traitées

	Vineyard/year				
	PARE 90	PULI 90	PULI 91	PULI 92-1	PULI 92-2
Number of treatments					
Fixed-interval	8	8	8	—	9
PLASMO	5	5	3	—	6
Disease intensity					
Fixed-interval	2.37 B	5.50 A	—	1.15 A	1.49 A
PLASMO	3.45 AB	7.29 A	—	0.29 A	1.95 A
Untreated	6.31 A	5.78 A	—	8.37 B	22.71 B

instance, delaying one application made it possible to cover two successive emergences of sporangia. This result could also be obtained using chemicals with a longer persistence. Fungicides for control of powdery mildew (*Uncinula necatrix*) were chosen among those not affecting downy-mildew development. Surface inorganic fungicides were applied in fixed-interval plots prior to flowering, while systemic fungicides were also used after fruit set.

In 1990, eight treatments were applied in fixed-interval plots, as against five applications recommended by the model in both Paretaio and Pulizzano vineyards. Disease intensity was determined at the end of the growing season (1990-09-10), before harvesting. The season did not present severe infection, as shown by the low value observed in untreated plots. In Pulizzano, no statistically significant differences ($P > 0.05$) were observed among the three treatments, while in Paretaio the fixed-interval plots showed a disease intensity significantly ($P < 0.05$) lower than in the untreated plots. PLASMO plots presented an intermediate value. The next year, 1991, there were no relevant downy-mildew attacks in untreated plots and thus DI was not determined. PLASMO identified four main infection cycles and only three fungicide application times. In fixed-interval plots, in contrast, eight treatments were applied. In 1992, prolonged and frequent rainfall during late spring and summer led to a high number of downy-mildew infections. However, the number of treatments applied in PLASMO plots was lower than in fixed-interval plots. Disease intensity was determined twice (1992-08-08 and 1992-09-28) with satisfactory results. No statistically significant differences ($P > 0.05$) were observed between fixed-interval and PLASMO plots, while the result for untreated plots was significantly higher ($P < 0.05$).

Conclusions

The model satisfactorily described grapevine downy-mildew development phases under field conditions. It was successfully applied in a grapevine protection programme, allowing a reduction in fungicide applications without increasing the severity of downy-mildew infection. PLASMO thus facilitates vineyard protection and could be considered a first step towards a global epidemiological model for grapevine protection against the main diseases and pests. It could be applied in the farm or in a computerized advisory system for plant protection. The global model could be transformed into an expert system (Saunders *et al.*, 1987) for grapevine management, based on the simulation of grapevine growth and development, and integrated with agronomical and phenological inputs. It could give useful information, especially when used in large areas, on phenological stage, treatment application times, fertilization, potential yield, economic threshold, etc.

In using this kind of models, particular attention should be paid to ensure that input data is representative of the area considered (vineyard, farm, district, territory, etc.). This is especially important in regions with high topographical variability, like the intensive and high-quality vine-growing areas of Italy.

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PLASMO: modèle de simulation pour la lutte contre le mildiou de la vigne

La protection de la vigne contre les maladies est un problème ancien, mais au cours de ces dernières années de nouvelles techniques ont été développées, visant à réduire les coûts pour l'agriculteur et les dégâts vis-à-vis de l'environnement. Elles sont basées sur des modèles mathématiques décrivant l'état du système plante-parasite-environnement. Un modèle de prévision du développement du mildiou de la vigne (*Plasmopara viticola*) est présenté dans cet article. Les variables d'entrée sont la température, la pluviosité et l'humectation foliaire (pour la contamination des sporanges), et l'humidité relative et la température (pour la période d'incubation). Simultanément, le modèle prend en compte la survie limitée des spores. La sortie du modèle s'exprime en termes de % d'avance de l'infection. Les tests de validation au champ, effectués en 1990, 1991 et 1992 dans plusieurs vignobles de la région Toscana (Italie centrale) ont montré une bonne corrélation entre les contaminations observées et simulées, permettant de réduire le nombre de traitements sans augmentation des dégâts dus au mildiou. Le modèle pourrait à l'avenir être intégré dans des modèles de simulation de la croissance et du développement de la vigne, afin de déterminer la surface contaminée et ainsi le seuil de nuisibilité économique.

PLASMO — имитационная модель для борьбы с *Plasmopara viticola* на винограднике

Защита виноградников от заболеваний не относится к числу новых проблем, однако за последние несколько лет были разработаны новые методы, позволяющие снизить затраты и уменьшить ущерб, наносимый экосистеме. Эти методы основаны на математических моделях, описывающих взаимосвязи между растением, паразитом и окружающей средой. В настоящей публикации приводится модель прогнозирования распространения ложной мучнистой росы виноградника (*Plasmopara viticola*). К вводимым переменным относятся температура, уровень дождевых осадков и увлажненность листа (признаки заражения спорангиями), а также относительная влажность и температура (для инкубационного периода). Модель также учитывает ограниченную выживаемость спор. Выходные данные получают в виде процентного выражения распространения заболевания. Тесты обоснования в природных условиях, проведенные в 1990, 1991 и 1992 годах на различных виноградниках Тоскании (Центральная Италия), продемонстрировали хорошее соответствие между наблюдаемыми иимитируемыми параметрами заражения. Модель позволяет снизить число обработок без увеличения поражения ложной мучнистой росой. В дальнейшем модель может быть объединена и вместе с моделями имитации роста и развития виноградников войти в экспертную систему для определения поверхности пораженной ткани и, следовательно, экономического порога ущерба.

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